Question 1

Luisa Kalkert, Luke Richard, Nicholas Chandler

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Introduction

In this section, we ask the question: "How do countries and continents differ with respect to selected human-geographic indicators and what conclusions could be drawn from these relationships?"

We focus on several geographic, demographic, and economic indicators from the CIA world fact book. Specifically, in our analysis of the aforementioned question, we examine indicators related to economic output, access to medical care, and societal demographics. To begin the analysis, grouped bar charts of arable land and total land per continent are examined. Thereafter, we examine boxplots of real GDP per capita, infant mortality rate, and total population for each country, grouped by continent. We then examine a corrplot of select numeric variables using both Pearson's and Spearman's correlation coefficient. To conclude the analysis of this questions, we examine PCA of the same numeric variables, visualizing a raw variance plot, a cumulative proportion of variance explained plot, a biplot, and a plot of scores where each score is color coded by continent and labeled with the corresponding country.

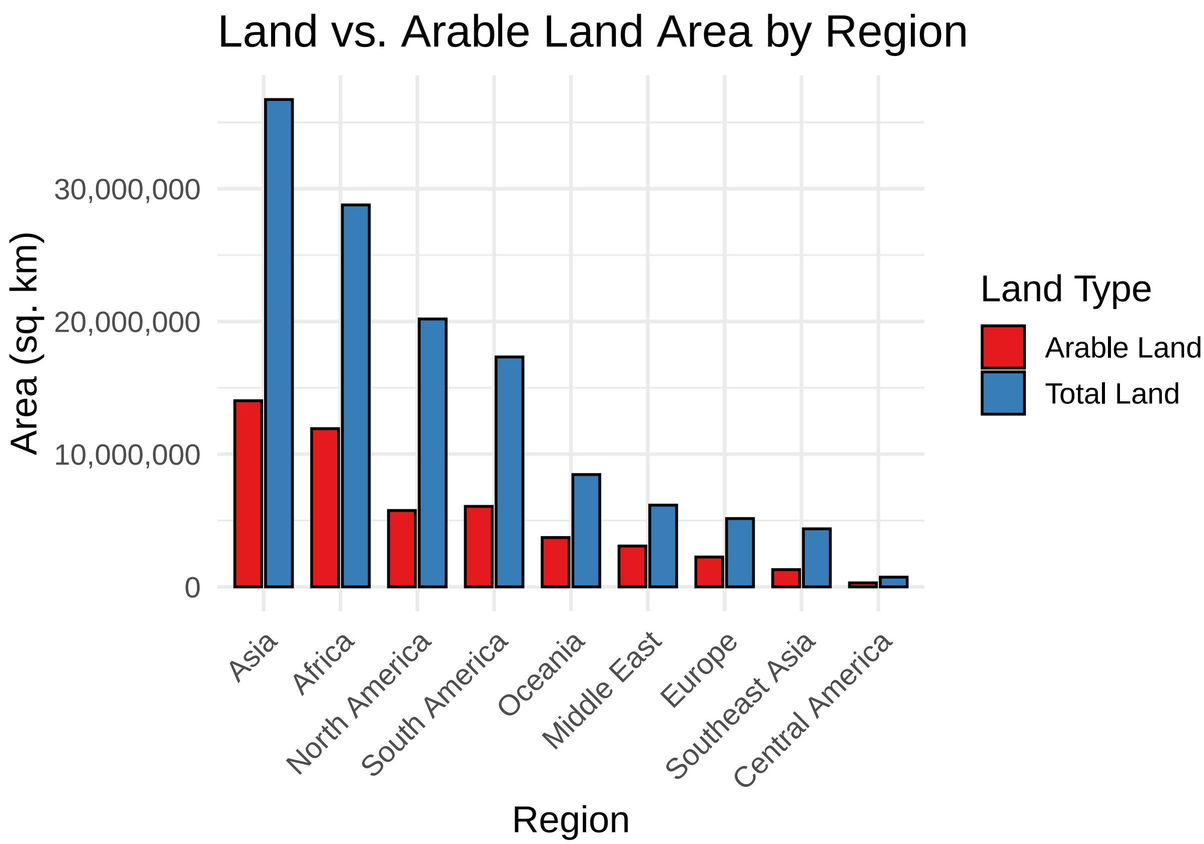
The continents/regions we examine are:

```
## [1] "Asia"            "Europe"          "Africa"          "Central America"
## [5] "South America"   "Oceania"         "Middle East"     "North America"
## [9] "Southeast Asia"
```

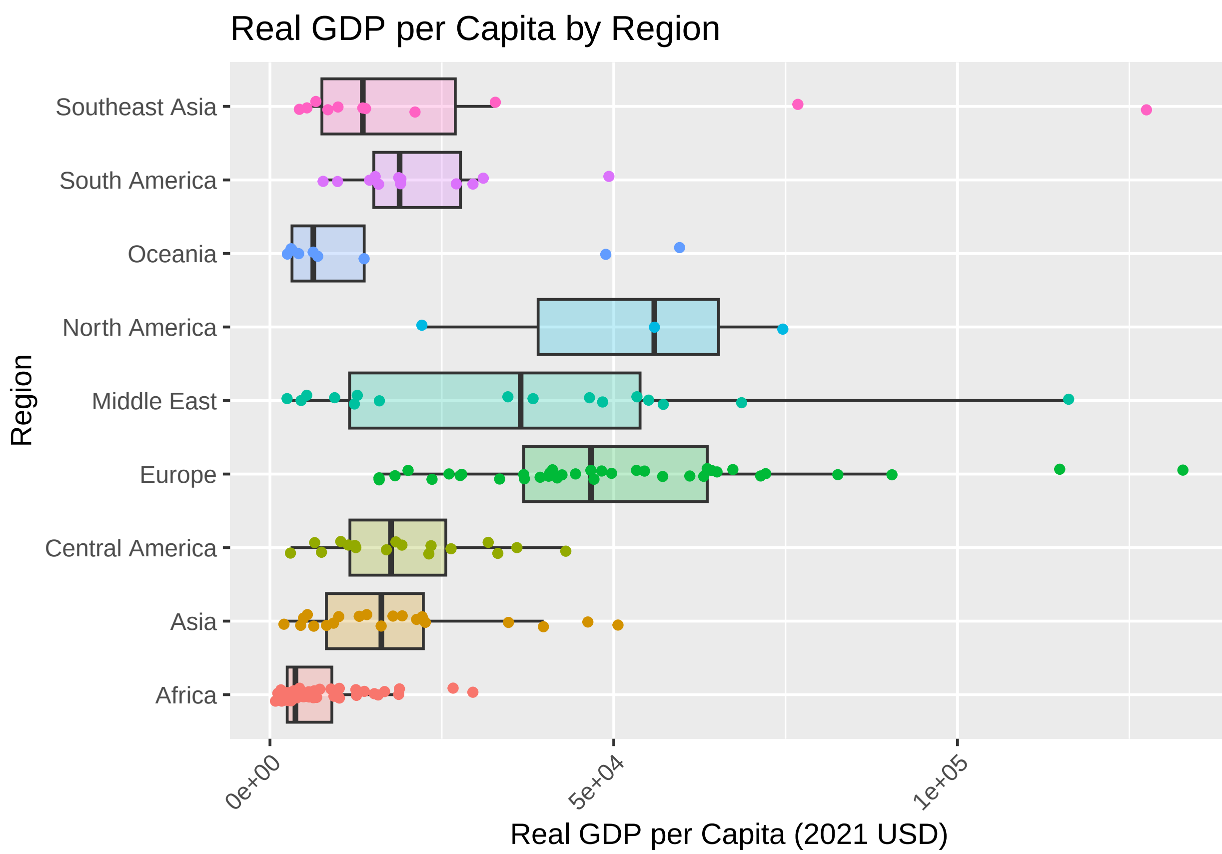
On the world map, these countries are: [world map with various countries color coded by region]

Exploratory Classical Graphics

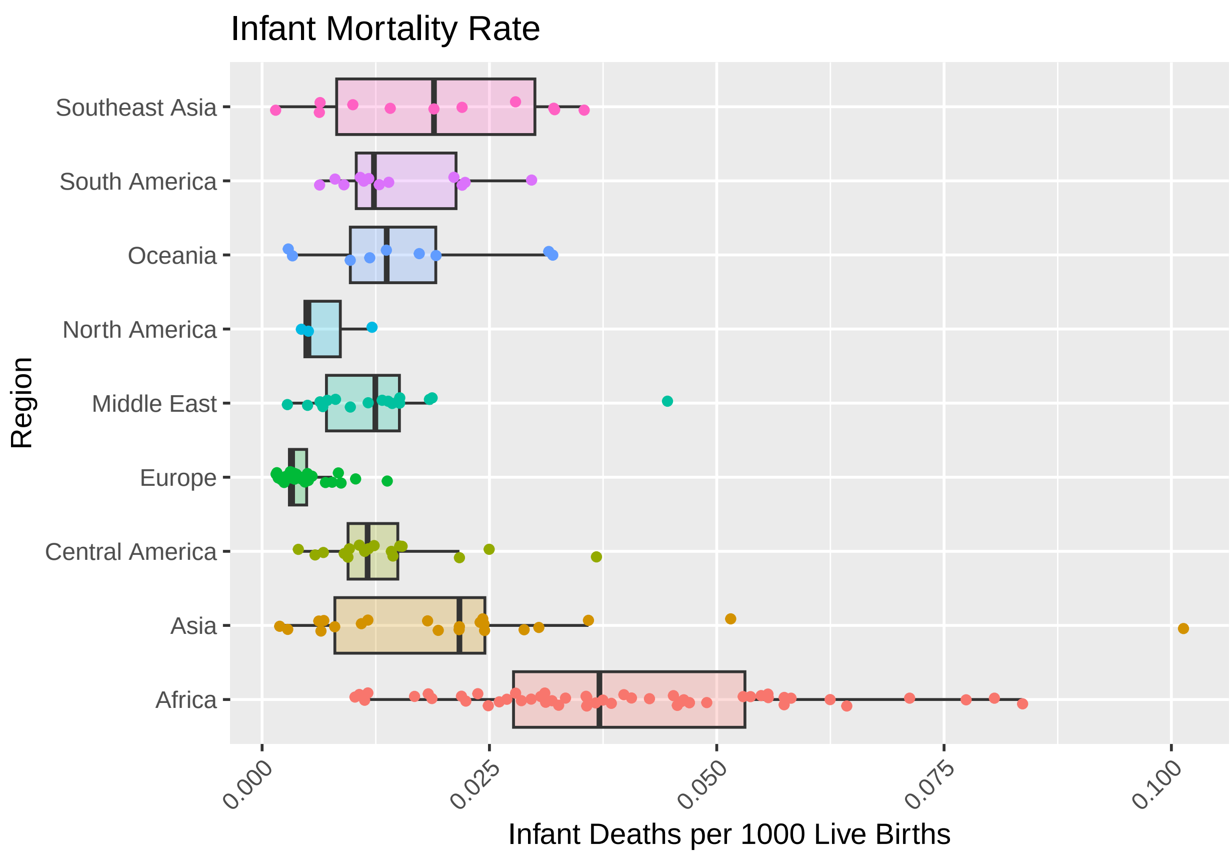
In this section, we examine a set of statistical graphics on a representative subset of the variables considered in the rest of this question. Specifically, we look at Land vs. Arable Land by region, GDP per capita, Infant Mortality Rate, and Total Population. The statistical graphics examined are grouped bar plots, and boxplots. These graphics yield a visual description of the distribution of the selected variables with respect to the region where each of the variables was measured.



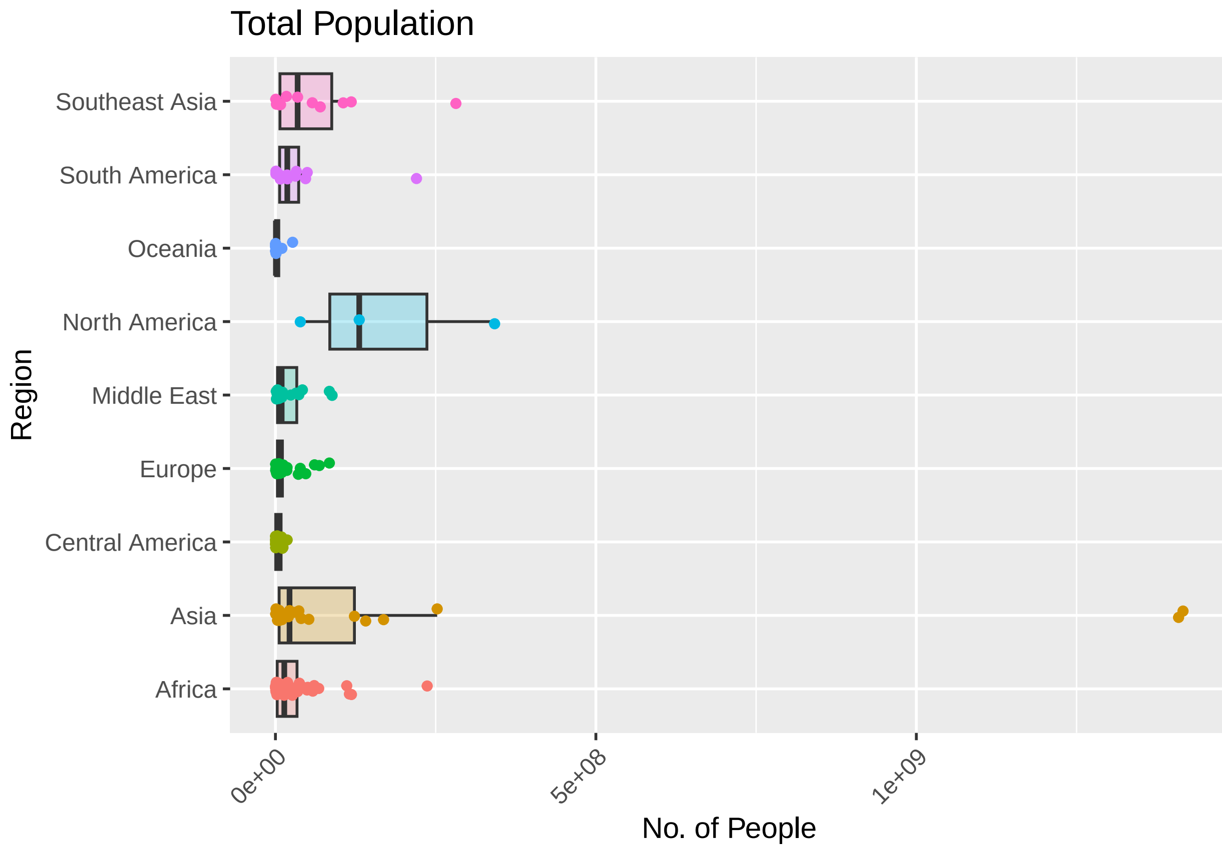
One can observe an ordering of the total land per region and arable land per region, though the ordering of each is not identical. We see that Asia has the most land and arable land while Central America has the least of both. We also see that South America has a higher proportion of arable land compared to North America.



We see that Africa has the lowest median real GDP per capita and North America has the highest. The Middle East has a very large spread compared to the other countries. The country with the highest real per capita GDP comes from Europe.



Africa has the highest median infant mortality rate while Europe has the lowest. Still, the country with the highest infant mortality rate comes from Asia, while the country with the lowest comes from Southeast Asia.

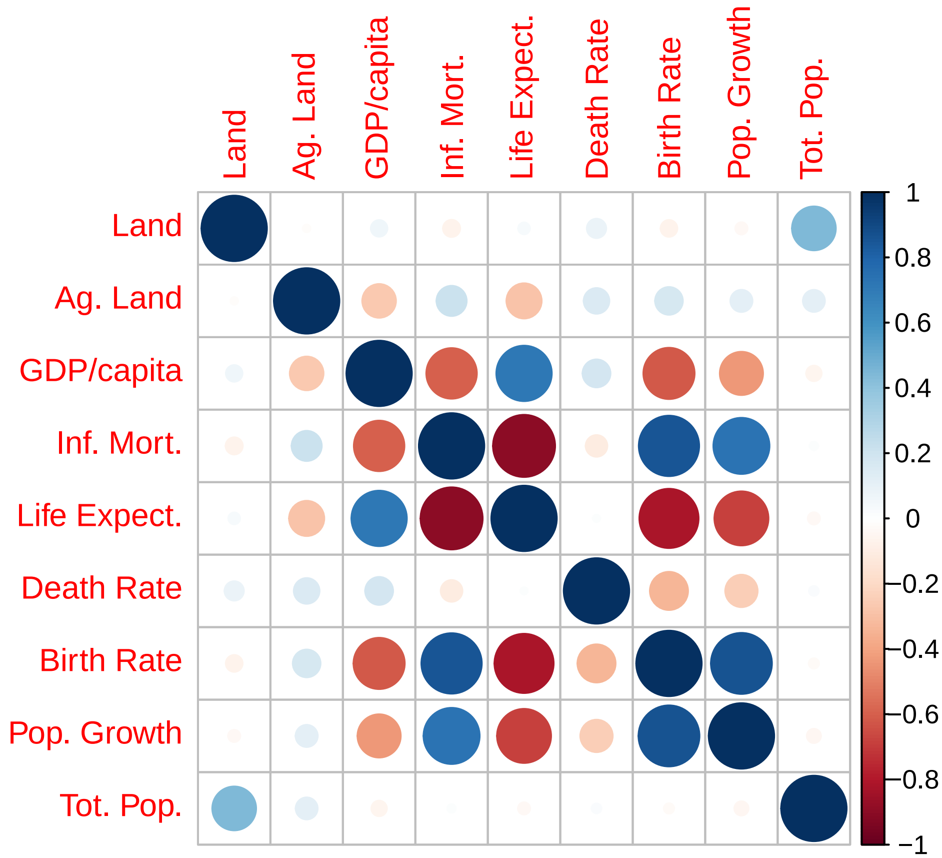


The region with the highest median total population is North America. North America also appears to have the highest IQR, a robust measure of dispersion. Asia has the two most populous countries in the world, India and China.

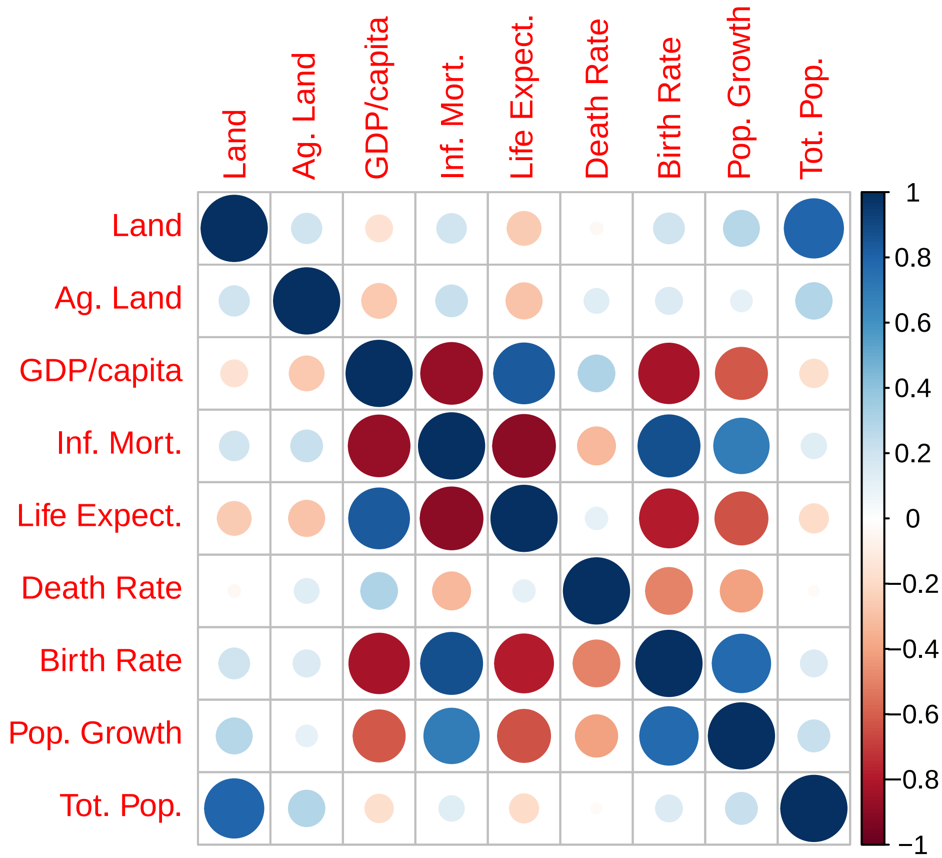
Correlation

In this section, we examine the correlation between selected numeric variables by means of a corrplot and pairs plots. The variables examined are:

```
## [1] "Total.sqkm"                        "Arable.land.percent"
## [3] "Real.GDP.per.capita.USD"           "Infant.deaths.per.1000.live.births"
## [5] "Life.expectancy.at.birth"          "Deaths.per.1000.people"
## [7] "Live.births.per.1000.people"       "Population.growth.rate.percent"
## [9] "Total.population"
```



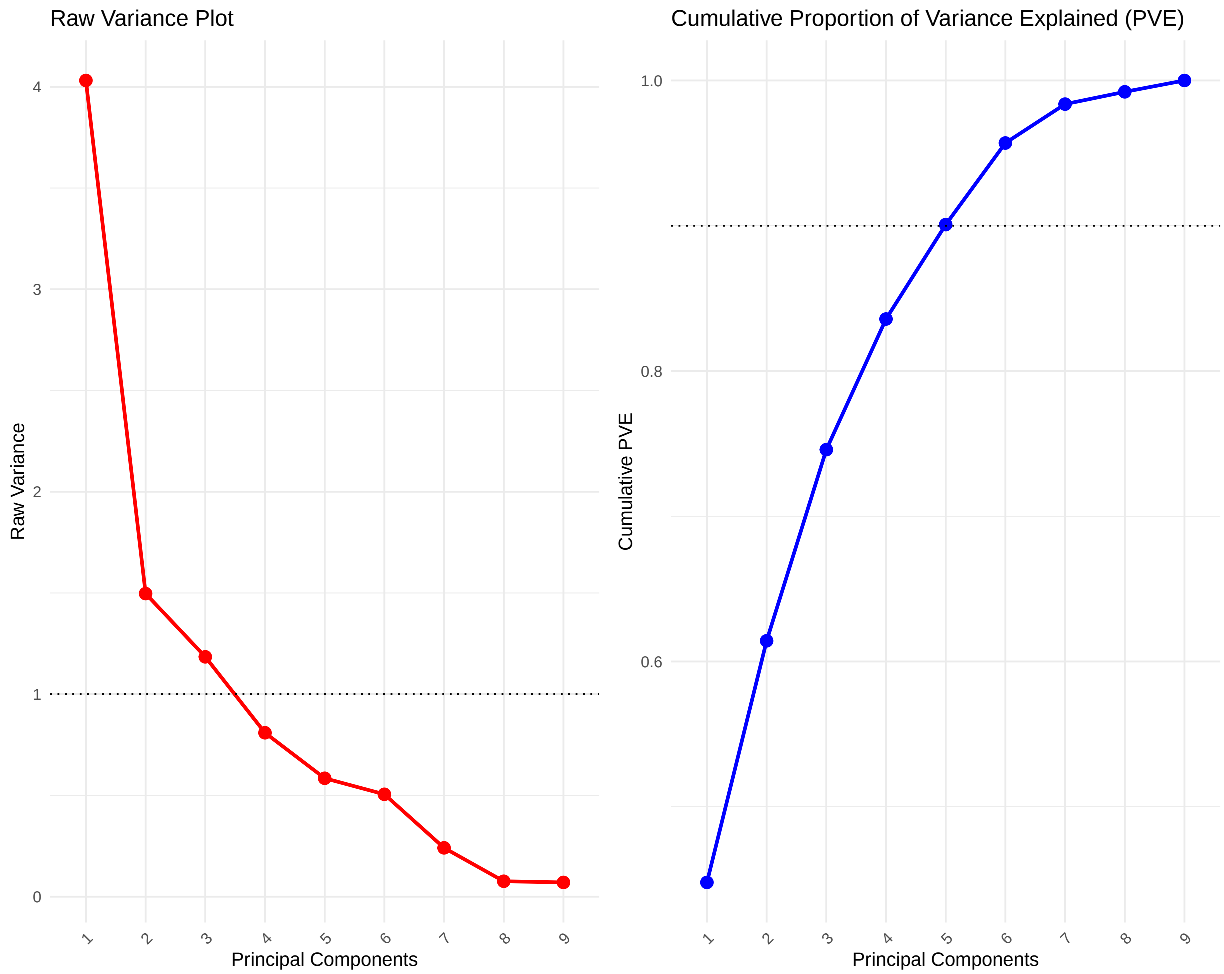
In this plot we see the Pearson correlation between the aforementioned numeric variables. Specifically, one can see that land, agricultural land, death rate, and total population have little correlation with other variables whereas the remaining variables seem to have complex correlation patterns among themselves. In this plot, one sees a strong negative correlation between life expectancy and infant mortality, a strong positive correlation between birth rate and population growth, a strong positive correlation between birth rate and infant mortality rate, and a strong negative correlation between birth rate and life expectancy. These correlations are linear and could perhaps be dependent on an underlying phenomenon such as access to medical care.



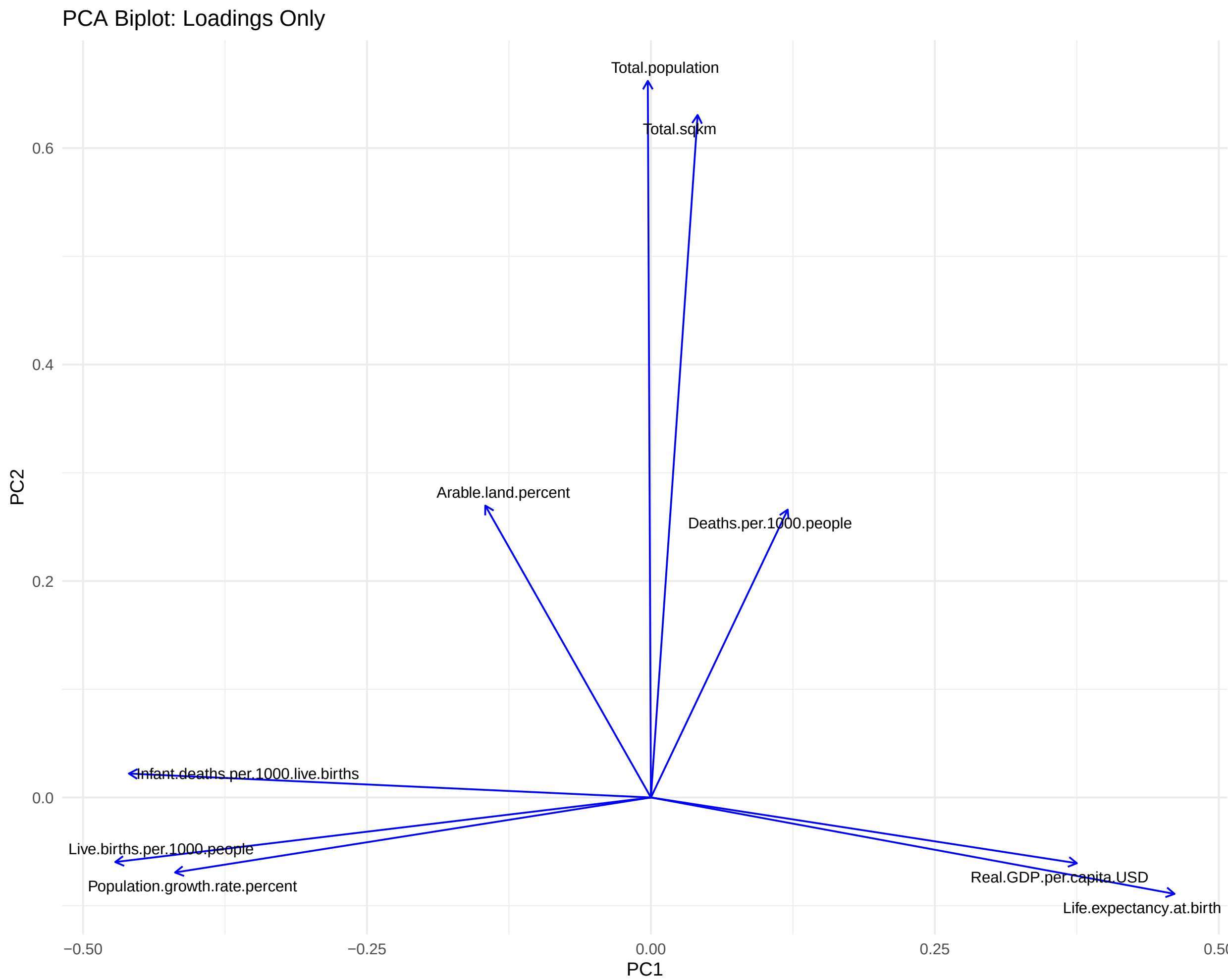
In this plot we see the Spearman correlation between the numeric variables. Although similar to the above plot, one can see higher positive correlation between land area and total population, suggesting that there is a monotonic relationship between the two variables though perhaps nonlinear. One can also see a stronger negative correlation between infant mortality and real GDP per captia, again suggesting that there is a strong, but non-linear relationship. There appears to be a weaker positive correlation between birth rate and population growth, perhaps suggesting that this is a linear relationship.

Principal Components Analysis

In this section we examine principal components analysis on the same variables as in `Correlation`. We begin the analysis by looking at the variance contained in each of the principal components, then we examine the principal component loading vectors and formulate interpretations of each of the first two principal components based on the loadings of each original variable. Finally, we look at the scores of the observations and examine per-continent trends based on our interpretation of the first two principal components.



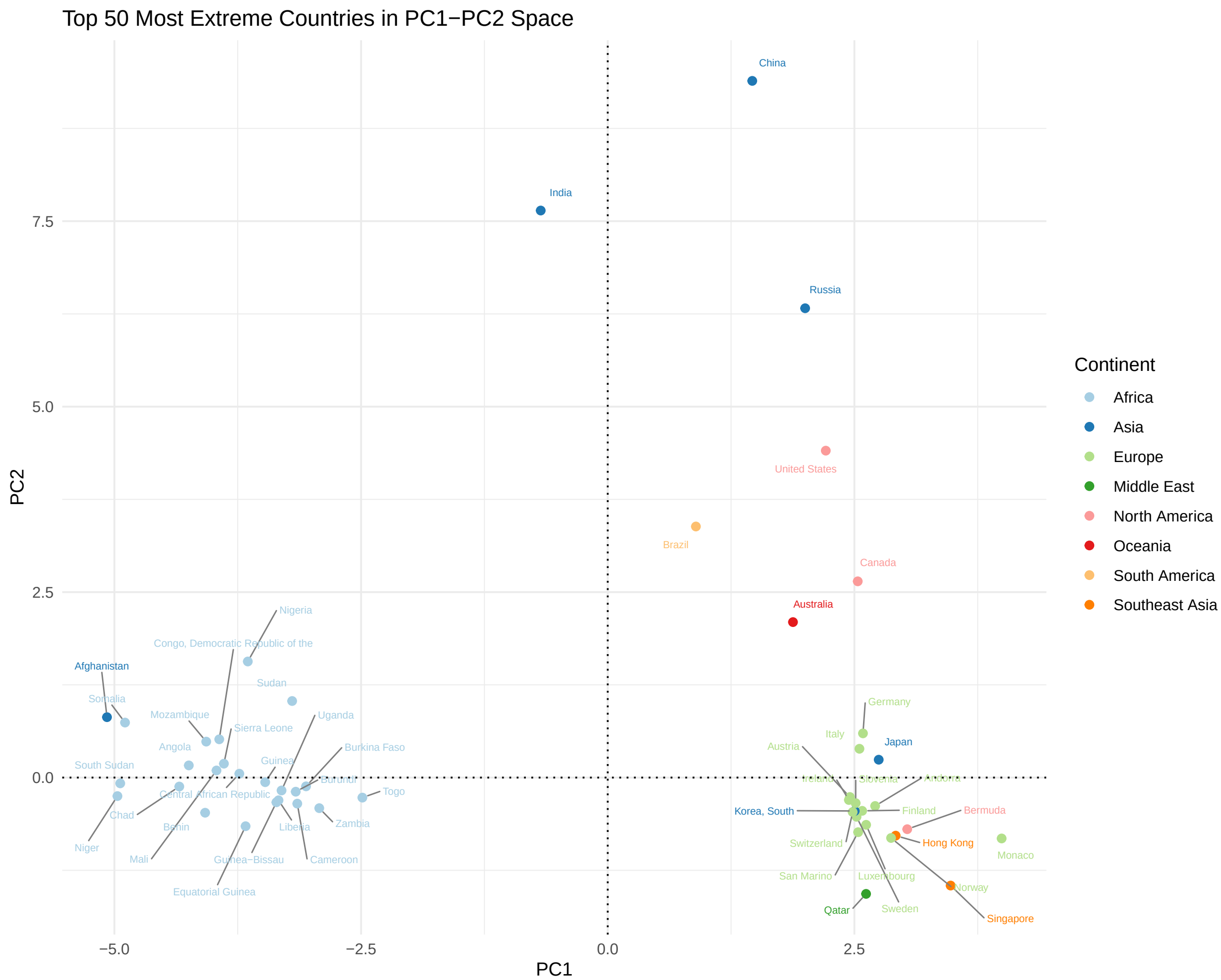
There are 3 principal components that "aggregate information" due to the variance being larger than 1 while the PCA was run on scaled data. Additionally, to capture ~90% of the variance, we need 5 PCs, and to capture ~75% of the variance, we need only 3 PCs.



In the above biplot of the principal component loading vectors we make the following observations:

One can see that for small values of PC1 we see large values of Infant Mortality Rate, Birth Rate, and Population Growth Rate. For large values of PC1 we see large values of Real GDP per capita USD and Life Expectancy at Birth. One could interpret PC1 as the developedness of countries with larger values corresponding to more developed countries.

For large values of PC2 we see that the countries typically have large values of Total Poplation and Total SqKm. To a lesser degree Arable Land and Death Rate also contribute to the PC2 direction. An interpretation of PC2 could be the size of a country in terms of population and area.



In the scores plot we plot the scores of the 50 countires farthest from the origin. We also make the following observations and conclusions:

We see groupings along the PC1 axis of more developed countries on the side with larger PC1 values and less developed countries with smaller PC1 values. There are several outliers with large values for PC2 that correspond with larger countries in terms of both population and land mass. Specifically, countries from Africa tend to have lower values of PC1 whereas countries from North America, Europe, and Southeast Asia tend to have larger values for PC1. Additionally, it appears that strong outliers account for the majority of the variance in PC2 as we see India, China, Russia, and the United States with the largest values whereas most other countries group near 0 with less variation.

Conclusion

Through this analysis, we have examined several variables with the aim of answering the question: "How do countries and continents differ with respect to selected human-geographic indicators and what conclusions could be drawn from these relationships?" Specifically, we looked at several classical graphics to gain a better understanding of how different continents relate with respect to a set of `high impact` human-geographic metrics. We then examined linear and monotonic correlation patterns between the different variables, indicating which variables had similarity. This investigation of the relationship between variables was continued with the PCA Biplot where we found that there were groupings that made the principal component axes interpretable. Finally, we examined the 50 most extreme countries in PC1-PC2 space and drew further conclusions about how the countries are related and differ with respect to the principal component axes.

The main takeaways are that there are only a few very large countries in the sense of total land and population, namely India, China, Russia, and the United States. Based on the variables examined, African countries tend to be less developed than Europe, North America, Southeast Asia, and Oceania. From this, one could speculate that the geographic location plays a large role in the developedness of nations as we see that there is grouping based on geographic location.